Input
FASTQ / FASTQ.GZ files or MinKNOW fastq_pass folder
(one sample per barcode)
Read filters
minimum read length (default 100 bp); optional minimum mean read Q
Orientation
reads oriented on the HTT sense strand (CAG vs CTG triplets):
both strands are used
Anchoring
5′ flank CTCAAGTCCTTC · start of the CCG repeat · CCT anchor
Read structure
(CAG)n | repeat-adjacent sequence | (CCG)m
CAG length = anchor-to-anchor distance / 3 (tolerant to isolated errors)
Configuration
canonical · CAA-loss · CCA-loss · CAACAGCCGCCA-loss ·
CAACAG-duplication (G2) · CAA-gain G1, G3–G8 · unresolved
Allele discovery
CAG modes searched within each configuration; each mode scored on
reads within ±1 CAG (±2 above 35); ≥5% of resolved reads, max. two alleles
Phasing
CCG length of each allele by majority of its reads; a single CAG mode is
split by CCG (two alleles of equal/adjacent CAG); chimeric reads ignored
Output
summary.tsv · result.json · haplotype counts · HTML report with
CAG, CAG×CCG, read-length and parsing-status plots